

R and Python

Overview of R

R is available for free, which is just one reason it is attractive for organizations of all types, but the huge number of tools it provides, in every aspect of data analysis, is at least as important.

- Use this link to open the webpage in an external window, for a brief introduction to R: [What is R?](#)

Once R itself has been installed, you will then get and also install an IDE, an Integrated Development Environment, which will provide a lot of extra tools to make using R much easier. A good IDE will be RStudio, one of the most commonly used environments for R.

- Use this link to learn about RStudio: [RStudio](#)

Summary of Python

Unlike R, Python is a computing language which can be used for a great many different purposes. While Python is also extremely popular for data analytics, it is also popular for many other development areas, including desktop applications, web programming, the Internet of Things, artificial intelligence, and much more. Open the following links to read a brief summary of the Python language [What is Python? Executive Summary](#)

Anaconda Distribution of Python

There are many versions of Python available, depending on what the user wants to do.

Anaconda provides a distribution (free download) of Python which includes a large number of **packages** which provide the code for data analytics, thereby saving users a lot of time which would otherwise be spent collecting and installing all the pieces needed.

Read a little bit about Anaconda here: [Anaconda Distribution](#) Anaconda will provide python.exe, which actually runs the code, plus all the other pieces needed to support the development of scripts for data analytics. This distribution will be installed first. Once Python itself has been installed, you will then get and also install an IDE, an Integrated Development Environment, which will provide a lot of extra tools to make using Python much easier. Some good IDEs are Visual Studio Code and Eclipse and LiClipse, which includes the specialty tools for Python, called PyDev.

[Among software developers, 'packages' refer to bundles of code, already written and ready for others to use, which are dedicated to specific tasks. For example, the '[nltk](#)' package includes many Python modules used for analysis of text.]

PyDev, LiClipse and RStudio

PyDev

Python itself is a general-purpose computing language. While there are many ways to run it, we will be using a platform which is designed specifically for Python users, called PyDev.

Most software developers use an Integrated Development Environment (IDE).

Eclipse is a very popular open-source IDE, which can be used with many different programming languages. Because Eclipse is **open-source**, meaning that the program source code itself is free, and can be modified, an organization called Brainwy has created a special edition of Eclipse called LiClipse, which includes PyDev already. This is a quick way to get started, since it saves users a lot of that would otherwise be needed to collect and install all the pieces needed individually.

- Learn a little bit about PyDev here: [What is PyDev?](#)
- Learn a little bit about LiClipse here: [LiClipse](#)

How to install R

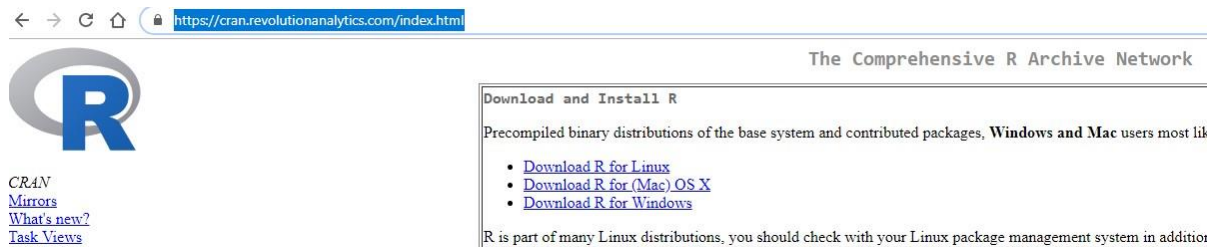
R is the actual statistical software itself; this comes first.

R is distributed by partners who support distribution by providing free copies at sites called mirrors.

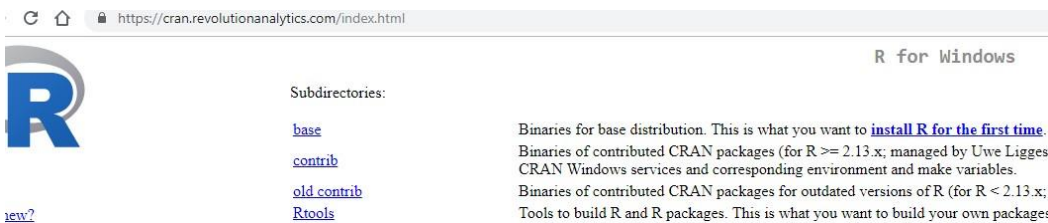
I obtained my latest copy from a mirror in the United States, in Texas: [The Comprehensive R Archive Network](#)

Instructions to Download R

1. Go to [The Comprehensive R Archive Network](#)
2. Since we're installing to your Windows platforms, use the "[Download R for Windows](#)" link.



3. The R for Windows page appears, select '**install R for the first time**'.



4. Finally we're at the download area! Click the link that says 'Download R 3.6.1 for Windows'.



At last, you're actually getting a download!

Mine was 82,697 kilobytes (about 83 megabytes), which only took a few seconds downloading on campus. This file is an executable, a program which will do the installation of R itself.

5. Click on the program to run the installer.
6. The wizard will run you through the standard questions.

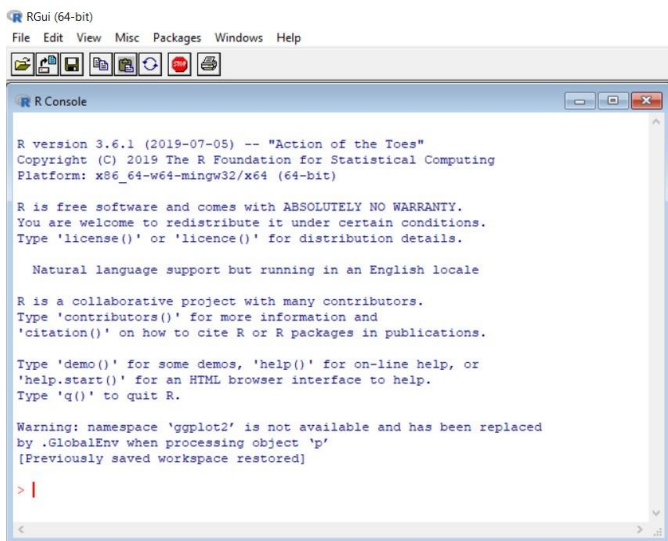
7. When it reaches the box for 'Select Additional Tasks', I like to 'create a desktop shortcut's, so I check that one as well. Otherwise, accept defaults.
8. Finish and exit the Setup wizard.

If you checked the box for a desktop shortcut, you should see this icon:



Double-click on the icon, and a simple **interface** opens.

[An **interface** in this context refers to the means by which a user controls the computer. As modern interfaces go, this one isn't very fancy: just a window with some menus, and a place to enter commands.]



The menus at the top supply a few standard functions. The primary tool here is the 'R Console'. The '>' symbol is the prompt where you can enter commands and see the results. For example, if you type the word 'version' at the prompt, R will report back version information about the software you just installed.

```
R Console
> version

platform      _
arch           x86_64
os            mingw32
system        x86_64, mingw32
status
major         3
minor         6.1
year          2019
month         07
day           05
svn rev       76782
language       R
version.string R version 3.6.1 (2019-07-05)
nickname      Action of the Toes
> |
```

That's it for R! Now that R is present, you can install RStudio.

How to Install RStudio

RStudio provides an Integrated Development Environment (**IDE**) that helps developers:

- work with R commands
- develop, test, debug, **run**, and save R scripts
- import and work with data
- view plots and visualizations
- install and work with packages

[To **run** software - a script, a program, an app - just means to execute the program instructions. IDE's provide multiple ways to run code.]

Instructions

1. Go to the website to download the FREE version of RStudio: [Choose Your Version of RStudio](#)
RStudio is available for multiple environments - different operating systems running different hardware. We will be working in a Windows / PC environment, so look for the 'Installers for Supported Platforms' section, and download the Windows installer. This installer supports Windows 7, 8 versions, and Windows 10. The version number you see may be different from the one in this screenshot from 8/12/2019.

Installers for Supported Platforms

Installers

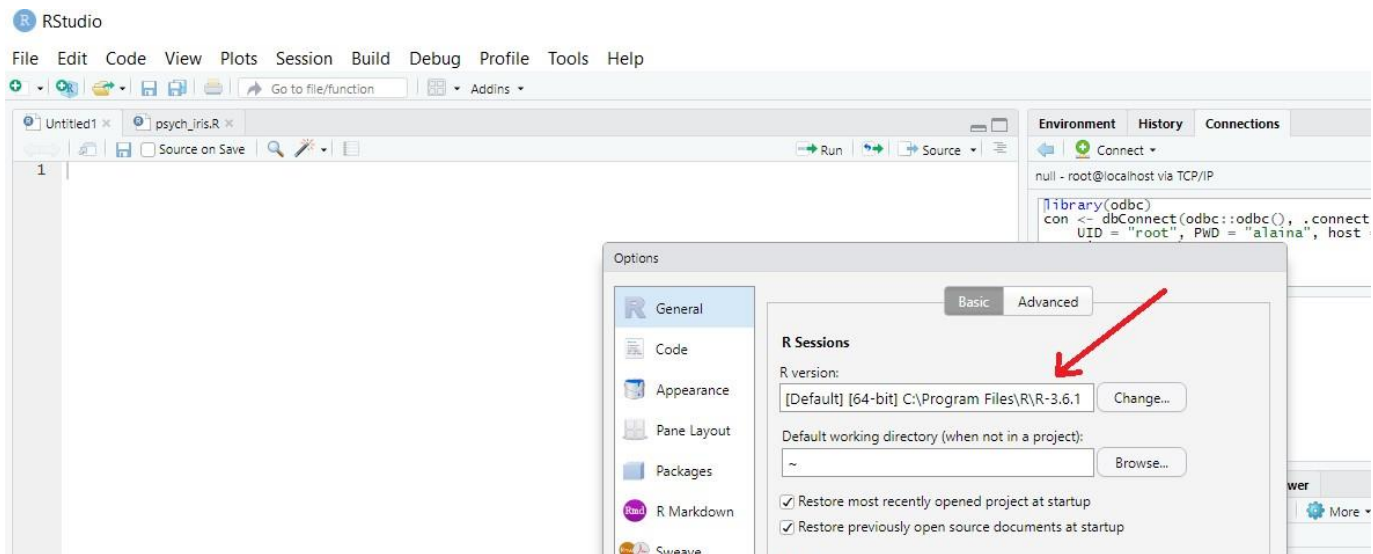
RStudio 1.2.1335 - Windows 7+ (64-bit)

2. Once the download is complete, double-click the .exe file to install.
3. The installer wizard opens - follow the steps, accept all the defaults.
4. Once all the steps are completed, click 'Finish' and close the wizard.

Tips for R and R Studio

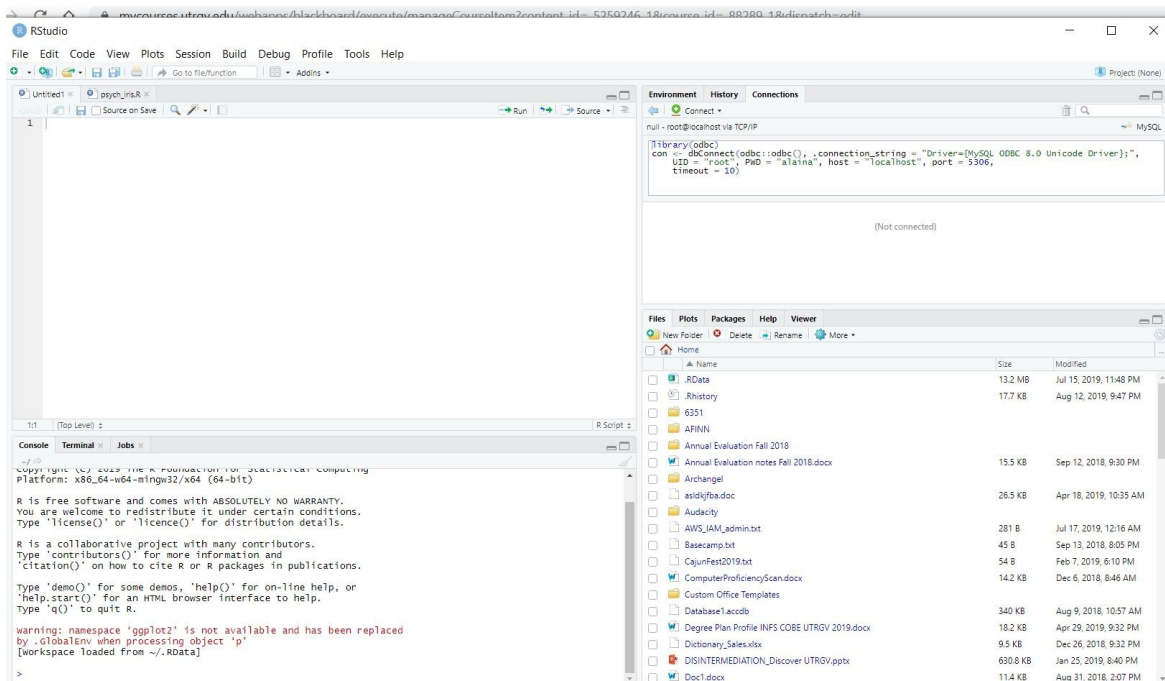
You can now find both R and RStudio in the Windows list of programs, or by typing 'R' or 'RStudio' in the Run window. I like to keep analytics software in the Taskbar. When the 'Run' window finds your software, you can put a permanent icon in the Taskbar by right-clicking the program name and using the 'Send to' command. RStudio finds your default installation of R when you install. You can check this by starting RStudio, then using the menu item

Tools - Global Options - General (on the left) - R version (on the right). This entry should reflect the R version you just installed.



I usually start R with ALL the panes showing. You can set this by using the menu:

View - Panes - Show all panes.



The script editor is on the upper left, the **console** (enter commands and see the results) on the lower left, and tabs providing more pane selections are on the upper and lower right.

The script editor is where we will work with commands in the R language. A sequence of commands (a script) will be saved, using the **file extension .R** (recall that Microsoft Word documents have the file extension .docx, programs have the file extension .exe, graphics have the extension .jpg or .png, etc).

Your script commands in R are handled by the file **R.exe**, which executes each line in your script. **R.exe** is already present on your computer, because it was installed in your first steps.

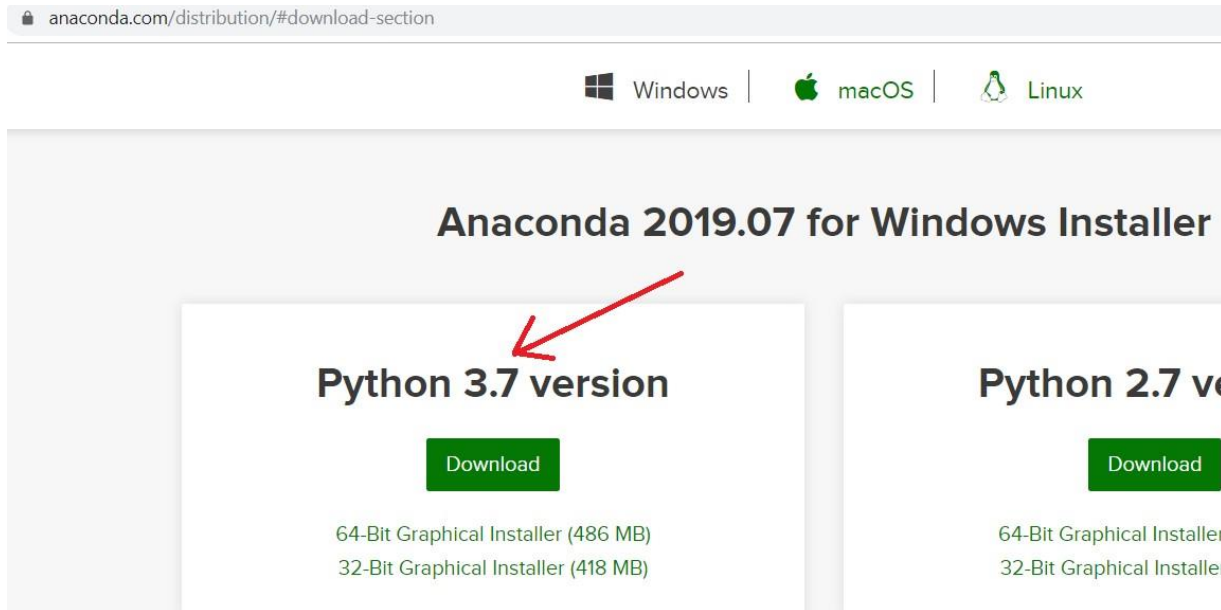
Installing Python with PyDev

Anaconda provides a large number of packages - software already written for Python which provide specific functions and tools - that are important for business analytics. [Anaconda Distribution Downloads](#)

NOTE: I have no idea how it works on MAC.

Instructions

1. Go to [Anaconda Distribution Downloads](#)
2. Download the executable installer from the website.



Anaconda provides extensive documentation and instructions. You can check your system requirements [Installation System Requirements](#)

- The **requirements** are the operating system and hardware you must have in order to be able to run the software.
 - Step-by-step installation instructions are available here: [Installing on Windows](#)
3. You may skip the Optional steps.
 4. The installation wizard will tell you where it is installing.
 - I recommend accepting the default - make sure you make a note of where it is installed, since we will use this information later to tell LiClipse where our preferred Python version is.
 - These instructions recommend that you DO make Anaconda your default Python - I agree, check this box.
 - They also recommend that you do NOT add Anaconda to your environment PATH variable - I agree, make sure this box is UNchecked.
 - Complete the steps and close the wizard. Anaconda should now be present in your Programs list.

You can check that Anaconda Python is installed and working by typing 'Anaconda Prompt' in the Windows 'Run' window.

This will open a command prompt window. IMPORTANT: This looks exactly like any Windows command prompt. The difference is that this one allows you to run Python immediately just by typing 'python' at the prompt. That command will be recognized, python.exe will start, and you will see the '>>>' python prompt.

```
Anaconda Prompt - python

(base) C:\Users\JeraldHughes>python
Python 3.6.4 |Anaconda, Inc.| (default, Jan 16 2018, 10:22:32) [MSC v.1900 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>
```

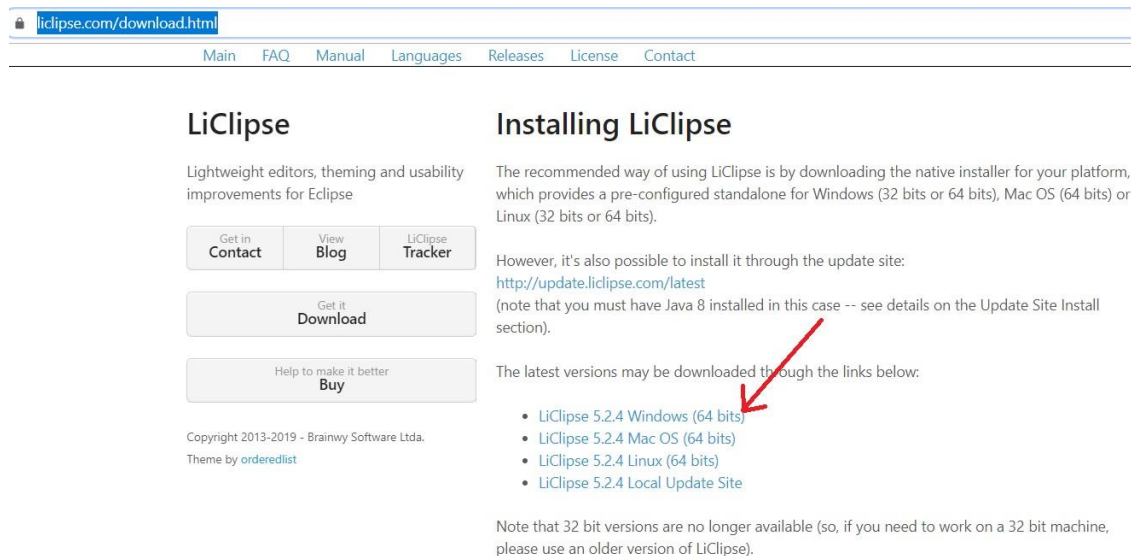
Installing Liclipse with PyDev

Now that Python itself is involved, you can install LiCclipse - this is an Integrated Development Environment (IDE) which comes with a bundled set of tools called PyDev, which are useful for developing Python applications.

LiCclipse is available here: [LiCclipse Download](#). LiCclipse is true shareware, meaning that the downloaded software is fully functional, but its creators will ask for a donation from those who use it.

Instructions

1. Go to [LiCclipse Download](#)



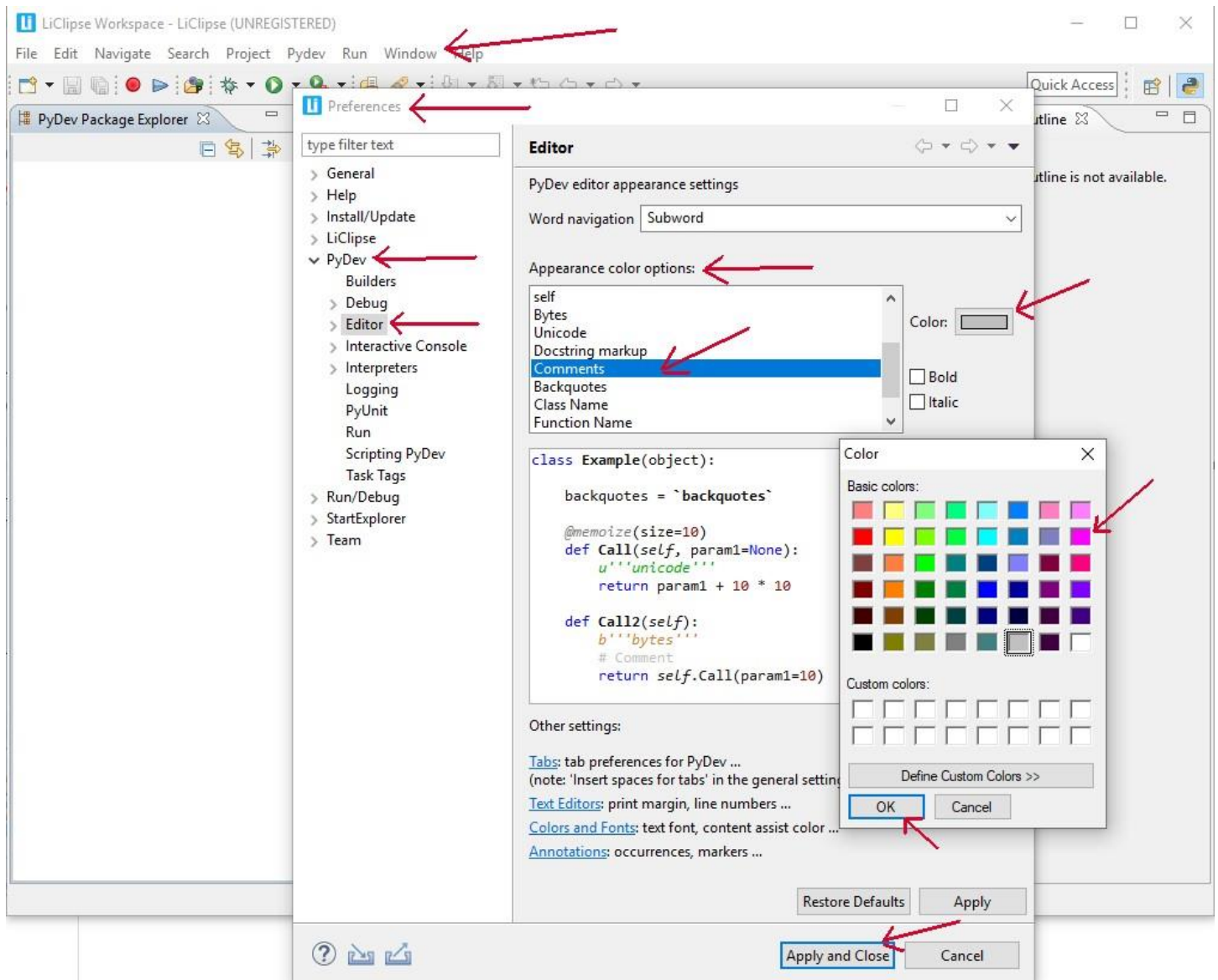
2. Click on the link to the Windows installer. The installer will download to your computer.
3. Click on the downloaded .exe file to install LiCclipse.
4. The installation wizard will start. There are only a few steps; accept all the defaults.
5. When the installation is done, start LiCclipse.

LiCclipse Interface

When LiCclipse opens, the launcher will ask you to select a workspace; this is where LiCclipse will store all your projects. Accept the default or create a folder yourself and browse to it. Next it will ask you to select a 'theme', which is just the appearance, meaning mostly the color choices. I still use the default, 'Eclipse Classic (Light Theme)', but the cool kids use dark themes nowadays. This one is up to you, just make sure I can read your code.

One thing I don't like about the Light Theme is the light gray comments, almost unreadable for my eyes. I have changed mine to bright pink. You can do this from:

Windows - Preferences - PyDev - Editor - Appearance Color Options - Comments - Color



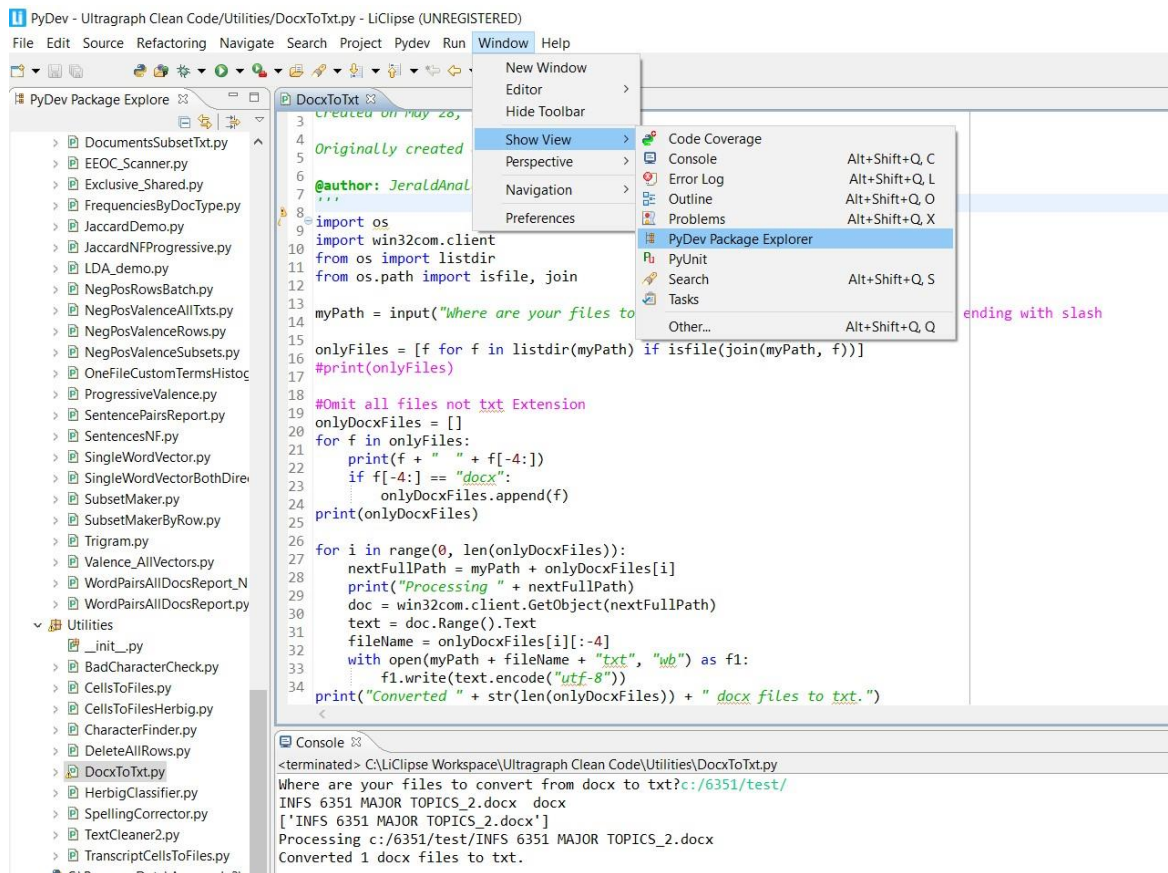
There are many ways to set up an IDE to display different information. For the purposes of INFS 6351, we will mainly want to see the PyDev Package Explorer. This is available from the menu item:

1. Select **Windows** from main menu
2. Select **Show View**
3. then select **PyDev Package Explorer**

In my latest install, this appeared already as part of the default layout (Yay!). We want to add one more pane we will use very often, the console. We can get this from:

1. Select **Windows** from main menu
2. Select **Show View**
3. then select **Console**

The following image shows my preferred layout, with the package explorer on the left, the **text editor** on the top right, and the console on the bottom right. The text editor area is likely the most important area, so I keep it as the largest pane on the screen.



Once you have a script, that is a Python **module**, which is a single file that contains the source code for a Python program. Each Python module is stored using the **file extension** **' .py '**.

LiClipse can use whatever interpreters you have available, and you can choose different ones for different projects.

Set LiClipse to identify Anaconda Python installation

In this next step, we will tell LiClipse where to find your Anaconda Python installation, so that you can choose this interpreter for your projects. Go to:

1. Select **Windows** from main menu
2. Select **Preferences**
3. Select **PyDev - Interpreters**
4. then select **Python Interpreter**

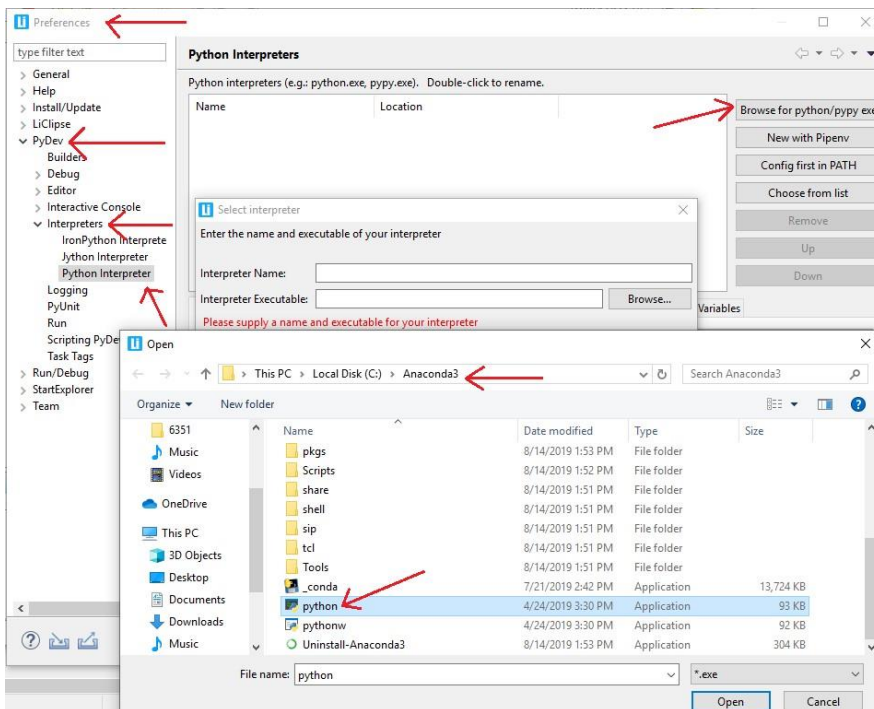
[COMPUTER SCIENCE-Y STUFF]

Nearly all modern computer languages work in part by re-using code that has already been written by someone else. Whenever the process you're running needs one of these pieces of code, it needs to know where it is. Your

Windows computer keeps track of all these pieces through a 'PATH variable'. This is just a list of where executable code is located: the drive and the folder(s) which store the code. Anaconda has a LOT of code to keep track of this way. If this list is not created, then when you try to run code, you'll just get a message saying that it's not found.

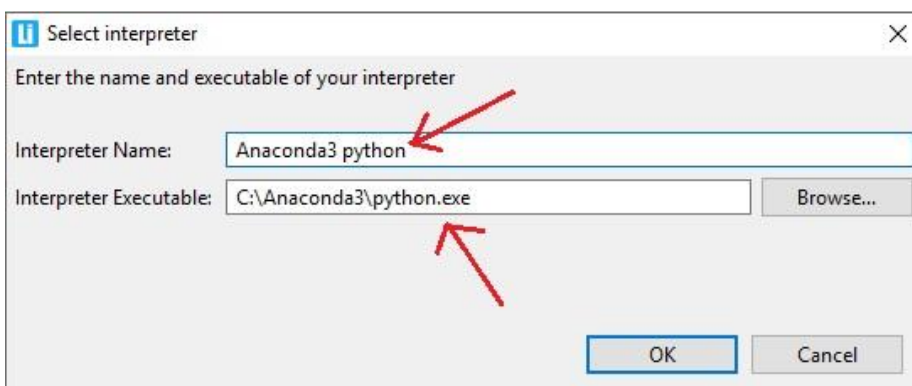
Here you can see the selection window that opens, to allow us to configure an interpreter. LiClipse wants to know where python.exe is. Remember in the installation of Anaconda3, you were instructed to "make sure you make a note of where it is installed" This is where you need it. When you click on 'Browse for python/pypy.exe', it is looking for that information. Notice that I created a directory directly under C: drive, and installed it there. If you accepted the default yours will be different--no problem, just use the popup browser box to where YOURS is located, and select that file.

When you click on the 'Open' button, AFTER you select the python file, the information will automatically be written into the 'Select Interpreter' box. Scroll down to the next graphic to see this.

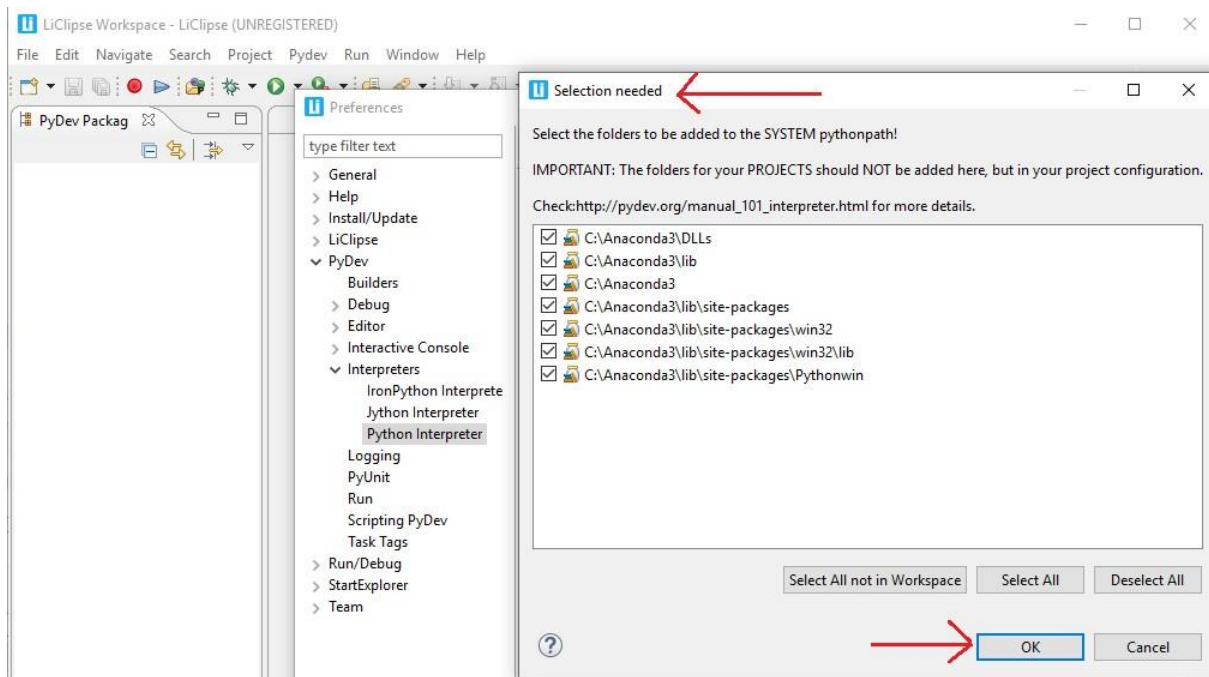


Here is an image of just the 'Select interpreter' box, with my information filled in. Remember, your entry here should be the actual location of YOUR python file.

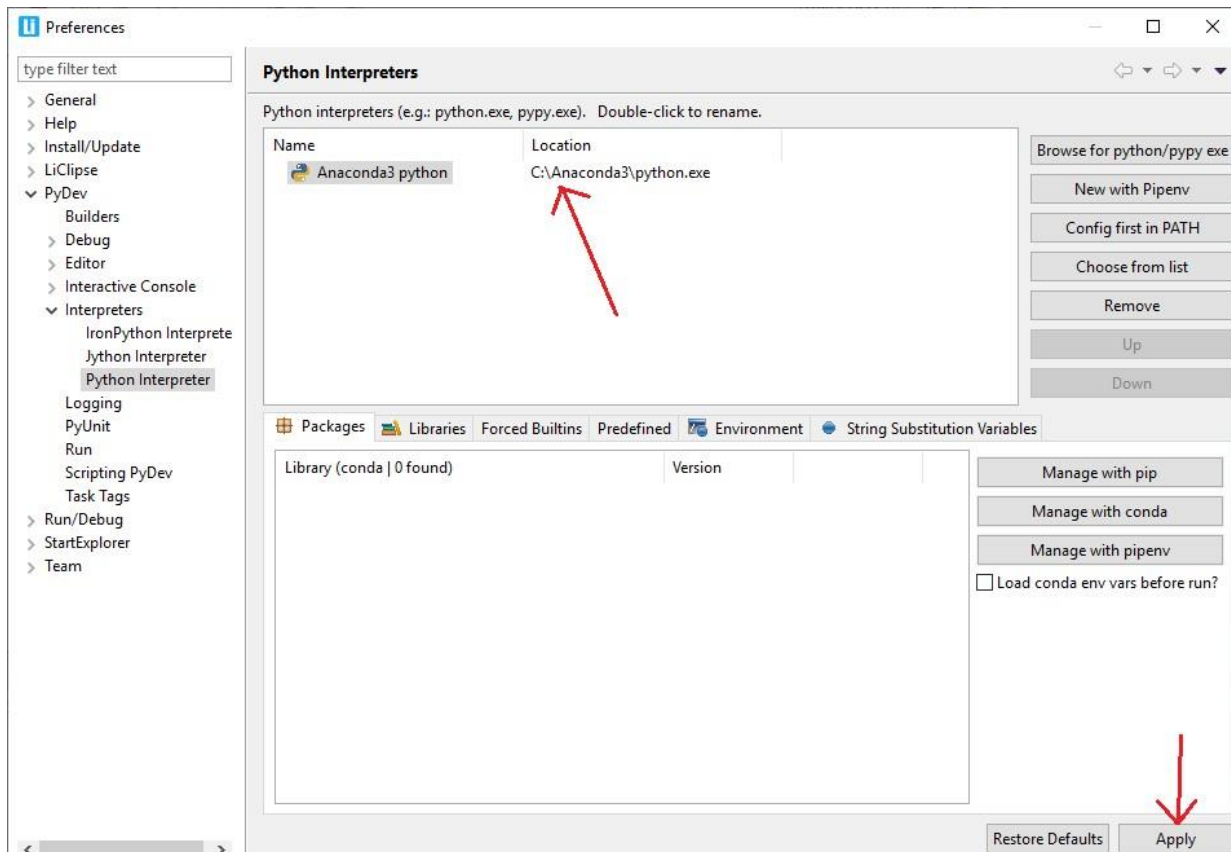
The default 'Interpreter Name' will be the name of the file, which is just 'python'. Notice that I have changed mine to Anaconda3 python. Name it whatever you like, I do recommend 'Anaconda3 python'. Hit 'OK'.



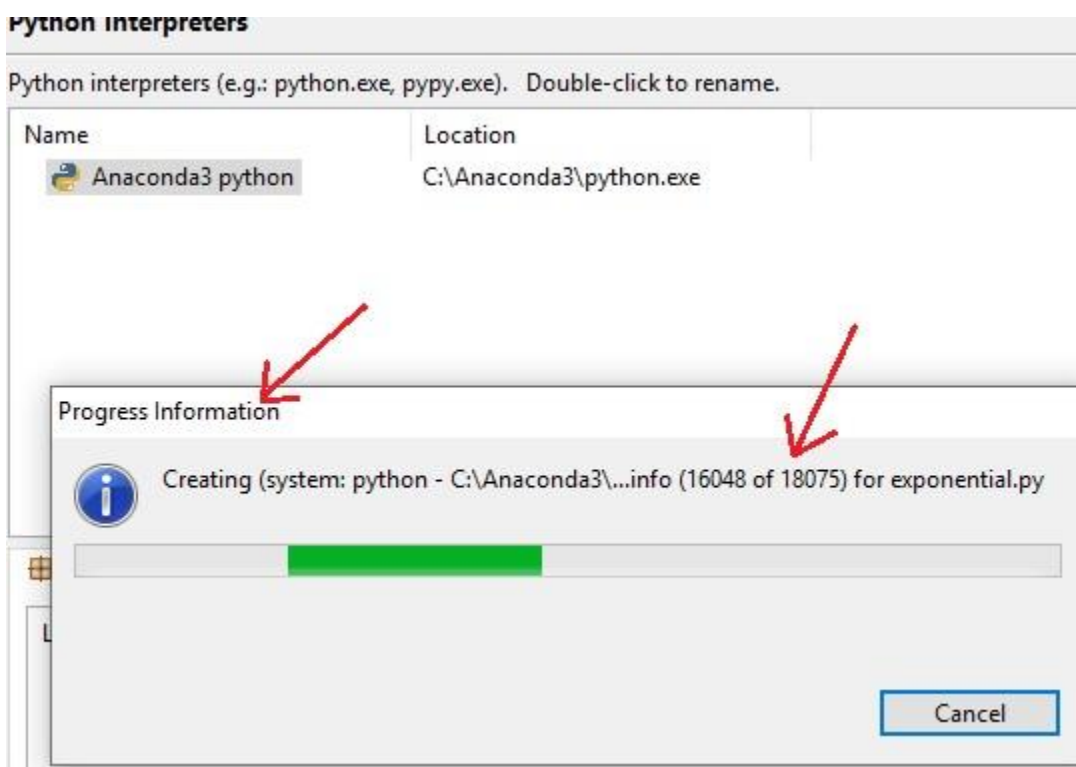
When you do this, LiClipse will realize that it needs more information, and a new box will open, 'Selection Needed'. THIS IS IMPORTANT. LICLIPSE CANNOT RUN YOUR CODE UNTIL ALL THESE STEPS ARE COMPLETE.



All the folders you need have already been found, so leave all the checkboxes marked exactly as they appear. Just hit 'OK'. It will show your single entry:



When you hit 'Apply', it will begin recording ALLLLLLL the location information. There was a LOT in my install, 18,075 entries. This will take a few minutes. Just watch the Progress bar and let it work until it's all done.



When it finishes, hit 'Apply and close', and the popup box will close, taking you back to the main LiClipse interface.

Now you are ready to do the first Python program, to show that your platform is installed and configured, ready to use.

Sources:

Anaconda (n.d.). Anaconda distribution. Retrieved from <https://docs.anaconda.com/anaconda/>

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Gordon, W. (2019, March 25). How to Run Windows on Mac. *PC Magazine* Retrieved from <https://www.pcmag.com/article/239012/how-to-run-windows-on-a-mac>.

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RStudio (n.d.). Choose Your Version of RStudio. Retrieved from <https://www.rstudio.com/products/rstudio/download/#download>